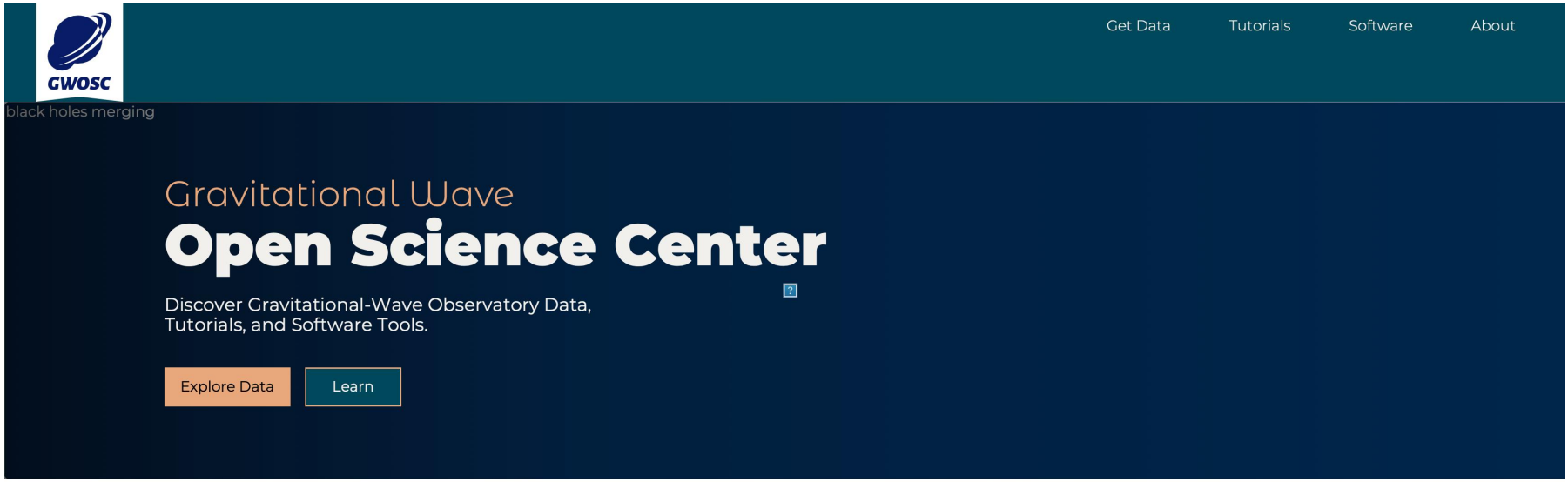
引力波参数估计简介

唐少鹏

2026-9-29

如何获得科学数据?



线性响应系统+反馈控制+标定（标定误差）

GW250114_082203

(1) O4_Discovery_Papers | (2) GWTC-5.0

Documentation

Release: GWTC-5.0

Event UID: GW250114_082203-v2

Names: GW250114_082203

GPS: 1420878141.2

UTC Time: 2025-01-14 08:22:03

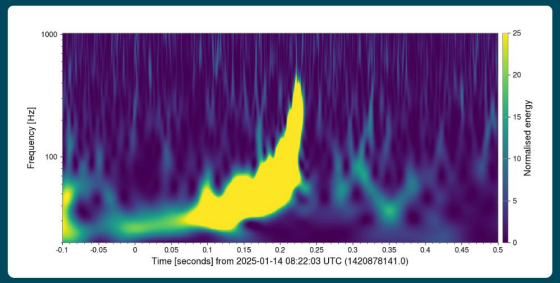
GraceDB: S250114ax

GCN: Query for Circulars · Notices

Timeline: Query for segments

DOI: https://doi.org/10.7935/bk00-6a89

H1 strain

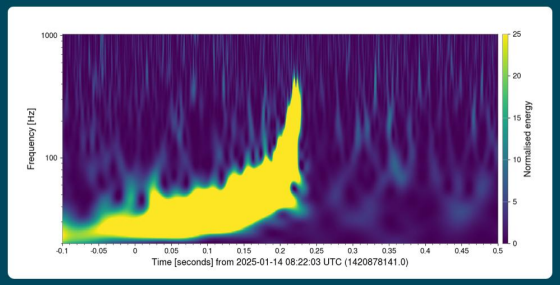


4096sec · 16KHz: HDF GWF

4096sec · 4KHz: HDF GWF

To open GWF files, use channels names as shown for GWTC-1:
https://doi.org/10.7935/82H3-HH23

L1 strain

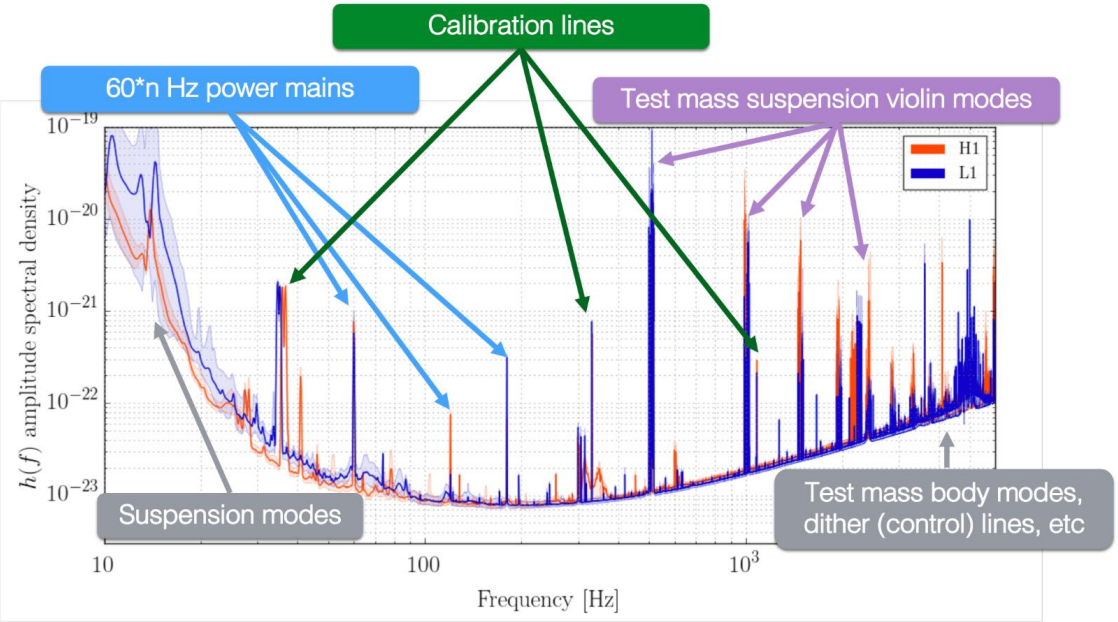


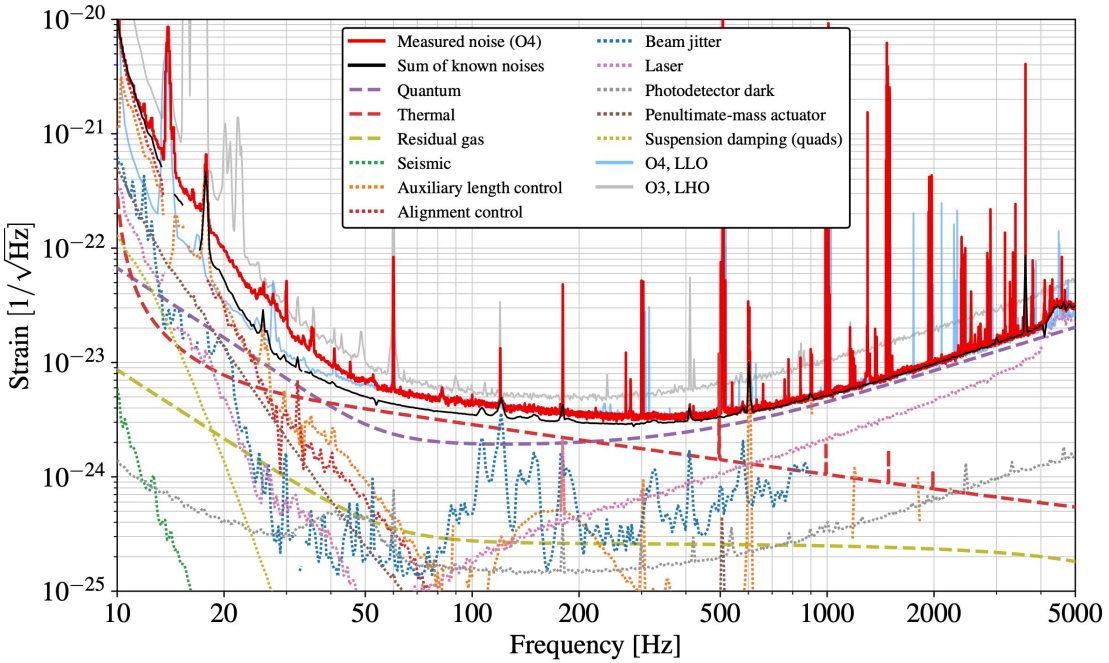
CWB Search Pipeline

Date added: May 22, 2026

show / hide parameters

已知噪声


Calibration lines
60*n Hz power mains
Test mass suspension violin modes
H1
L1
$h(f)$ amplitude spectral density
10^{-19}
10^{-20}
10^{-21}
10^{-22}
10^{-23}
Suspension modes
Test mass body modes, dither (control) lines, etc
10
100
10^3
Frequency [Hz]


Measured noise (O4)
Sum of known noises
Quantum
Thermal
Residual gas
Seismic
Auxiliary length control
Alignment control
Beam jitter
Laser
Photodetector dark
Penultimate-mass actuator
Suspension damping (quads)
O4, LLO
O3, LHO
Strain [$1/\sqrt{\text{Hz}}$]
10^{-20}
10^{-21}
10^{-22}
10^{-23}
10^{-24}
10^{-25}
10
20
50
100
200
500
1000
2000
5000
Frequency [Hz]

未知噪声

Gravity Spy 2.0 Wiki

Search...

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- Fast Scattering vs Slow Scatt...
- ER & O

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TALK

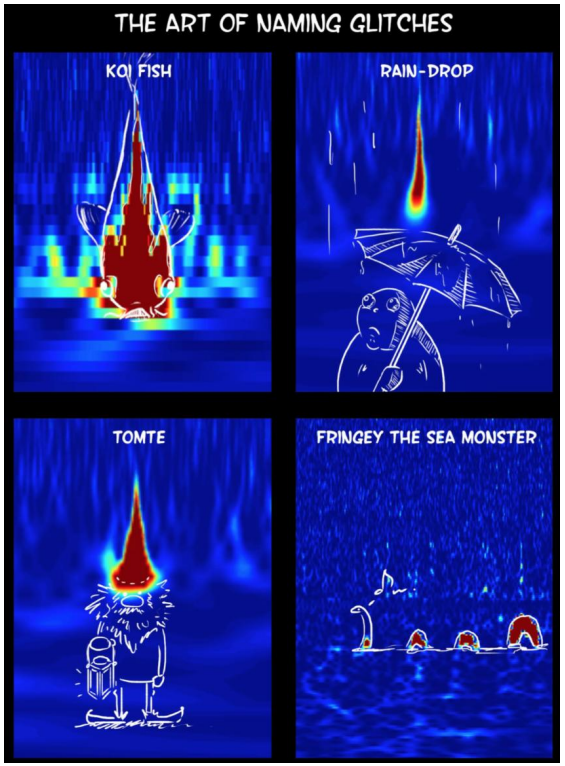
View Discussion

LAST EDITED BY
MarleneTA
09/30/2023

There are a couple dozen types of glitches that we currently know of. Over time, some glitches have come and gone in the detectors as their causes have been identified (such as an Air Compressor glitch, which was in the data from the first observing run but should not be around anymore as data from future observing runs rolls in). Some glitches have remained a mystery and persisted in the detectors (such as Blip glitches). As of now, the Gravity Spy project has more than persisted20 different options for glitch morphologies, which are described in the pages in this Wiki.

List of LIGO Glitches

- 1080 Line
- 1400 Ripple
- 70 Hz Line
- Air Compressor (50 Hz)
- Blip
- Chirp
- Crown
- Extremely Loud
- Helix
- Koi Fish
- Low-Frequency Line
- No Glitch
- Paired Doves
- Pizzicato
- Power Line (60 Hz)
- Repeating Blips
- Scattered Light
- Scratchy
- Tomte
- Uncategorized_Glitches
- Violin Mode Harmonic
- Wandering Line
- Whistle



基本概念

引力波探测器经过标定后的数据通常写成

d(t)=h(t)+n(t)　[有信号时]

无法知道某一时刻的噪声具体是多少，只能描述其统计规律!!! 需把**n(t)**看成一个随机过程。

什么是随机过程？ 由时间或其他参数标记的一族随机变量。

随机变量

一次掷骰子的结果 X是一个随机变量。

随机向量

连续掷三次骰子：(X1,X2,X3)构成随机向量。

随机过程

如果每个时刻都有一个随机变量{n(t):t∈R}就称为随机过程。

什么是realization? 实验中我们通常只有一条探测器数据，无法重复制造无数个完全相同的宇宙。

平稳（高斯）随机过程

是么叫平稳？　统计规律不随时间平移而改变（一般只考虑宽平稳）

高斯动机：中心极限定理+线性响应系统

$$t_1, t_2, \ldots, t_N,$$

$$\boldsymbol{n} = \begin{pmatrix} n(t_1) \\ n(t_2) \\ \vdots \\ n(t_N) \end{pmatrix}$$

$$p(\boldsymbol{n}) = \frac{1}{\sqrt{(2\pi)^N \det \mathbf{C}}} \exp\left[-\frac{1}{2}(\boldsymbol{n}-\boldsymbol{\mu})^{\mathsf{T}} \mathbf{C}^{-1} (\boldsymbol{n}-\boldsymbol{\mu})\right],$$

$$\mu_i = \mathbb{E}[n(t_i)]$$

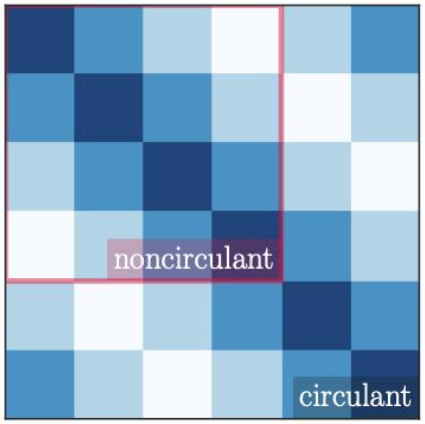
$$C_{ij} = \mathbb{E}\left[(n(t_i)-\mu_i)(n(t_j)-\mu_j)\right]$$

Wiener-Khinchin定理　$\frac{1}{2} S_n(f) \equiv \int_{-\infty}^{+\infty} \mathrm{d}\tau \rho(\tau) \mathrm{e}^{-\mathrm{i}2\pi f \tau}$，　自协方差函数

$R(\tau)$ 连续且 $\int |R(\tau)| \, \mathrm{d}\tau < +\infty$　遍态历经：时间平均替代系综平均。

Toeplitz矩阵

$$p(\mathbf{d}|\theta) = \frac{1}{\sqrt{(2\pi)^N \det \mathbf{C}}} \exp\left[-\frac{1}{2}(\mathbf{d}-\mathbf{h})^\dagger \mathbf{C}^{-1}(\mathbf{d}-\mathbf{h})\right].$$



SYMMETRIC
NONCIRCULANT
TOEPLITZ MATRIX

$$\begin{pmatrix} \rho_0 & \rho_1 & \cdots & \rho_{N-1} \\ \rho_1 & \rho_0 & \cdots & \rho_{N-2} \\ \vdots & & \ddots & \vdots \\ \rho_{N-1} & \cdots & \rho_1 & \rho_0 \end{pmatrix}$$

$$\mathbf{C}_{\mathrm{T}}^{(8)} = \begin{pmatrix} 4 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 2 & 4 & 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 4 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 4 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 4 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 4 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 2 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 4 \end{pmatrix}. \qquad \mathbf{C}_{\mathrm{circ}}^{(8)} = \begin{pmatrix} 4 & 2 & 1 & 0 & 0 & 0 & 1 & 2 \\ 2 & 4 & 2 & 1 & 0 & 0 & 0 & 1 \\ 1 & 2 & 4 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 4 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 & 4 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 4 & 2 & 1 \\ 1 & 0 & 0 & 0 & 1 & 2 & 4 & 2 \\ 2 & 1 & 0 & 0 & 0 & 1 & 2 & 4 \end{pmatrix}.$$

循环近似：可用FFT对角化得到频域PSD $\tilde{C}_{ij} \equiv E[\tilde{n}_i \tilde{n}_j] = \frac{1}{2} T\, S_n(f_i)\, \delta_{ij}$,

时域分析 vs 频域分析

时域与频域原则上是同一个统计模型的两种表示；
只有在作频域对角近似时，二者才可能出现差异。

$$p(\boldsymbol{n}) = \exp\left\{-\frac{1}{2}\left[(\boldsymbol{n} \mid \boldsymbol{n}) + \int \ln S_n(f)\,\mathrm{d}f + \text{const.}\right]\right\}$$

$$(\boldsymbol{a} \mid \boldsymbol{b}) = 2\int_0^\infty \frac{\tilde{a}(f)\tilde{b}^*(f) + \tilde{a}^*(f)\tilde{b}(f)}{S_n(f)}\,\mathrm{d}f$$

$$= 4\,\mathrm{Re}\int_0^\infty \frac{\tilde{a}^*(f)\tilde{b}(f)}{S_n(f)}\,\mathrm{d}f\ .$$

$$p(\boldsymbol{n}) = \exp\left\{\sum_i -\frac{1}{2}\left[\frac{4|\tilde{n}_i|^2}{TS_n(f_i)} + \ln\left(\pi T S_n(f_i)/2\right)\right]\right\}\ .$$

混叠与泄漏效应

混叠效应：采样定理,频谱失真，降采样一般必须低通。

泄漏效应：有限数据长度

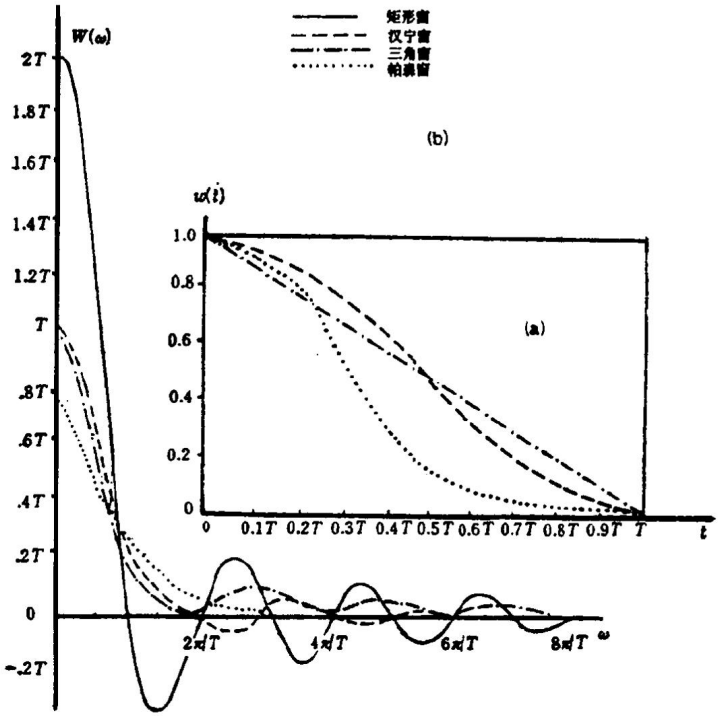


图 6.4　几种常用的平滑窗
(a) 时窗　(b) 谱窗

例　讨论信号 $y(t)=\cos\pi at$ 在 $|t|\leqslant T$ 内的频谱.

解　对无限长信号 $y(t)$，由(4.36)式知

$$Y(\omega)=\pi[\delta(\omega-\pi a)+\delta(\omega+\pi a)],$$

而截断信号 $x(t)$的频谱

$$X(\omega)=\frac{1}{2\pi}Y(\omega)*W(\omega)$$

$$=\frac{1}{2}\left[\delta(\omega-\pi a)*\frac{2\sin\omega T}{\omega}+\delta(\omega+\pi a)*\frac{2\sin\omega T}{\omega}\right]$$

$$=\frac{\sin(\omega-\pi a)T}{\omega-\pi a}+\frac{\sin(\omega+\pi a)T}{\omega+\pi a}.$$

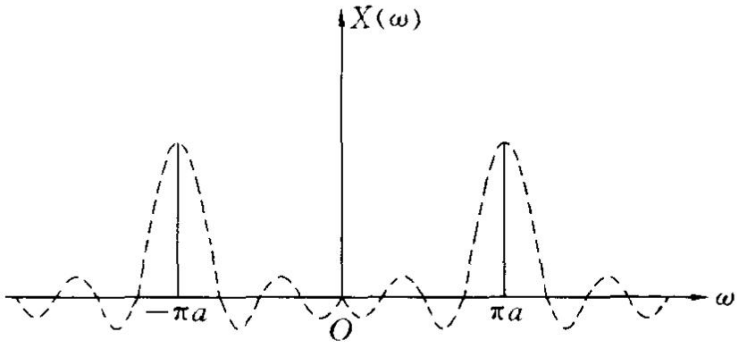


图 4.10　信号的泄漏效应

似然函数比

贝叶斯公式： $p(\boldsymbol{\vartheta} \mid \boldsymbol{d}, \mathcal{H}) = \dfrac{p(\boldsymbol{\vartheta} \mid \mathcal{H})\, p(\boldsymbol{d} \mid \boldsymbol{\vartheta}, \mathcal{H})}{Z}$,

模型比较： $\mathcal{O}_{ij} = \dfrac{p(\mathcal{H}_i \mid \boldsymbol{d})}{p(\mathcal{H}_j \mid \boldsymbol{d})} = \dfrac{p(\mathcal{H}_i)}{p(\mathcal{H}_j)} \times \dfrac{Z_i}{Z_j}$,

完备互斥假设： $\mathcal{H}_0 : \quad d = n, \quad \mathcal{H}_1 : \quad d = h(\boldsymbol{\theta}) + n,$

$$p(\mathcal{H}_1|d) = \left[1 + \frac{p(\mathcal{H}_0)}{p(\mathcal{H}_1)} \frac{p(d|\mathcal{H}_0)}{\int d\boldsymbol{\theta}\, p(\boldsymbol{\theta}|\mathcal{H}_1) p(d|\boldsymbol{\theta}, \mathcal{H}_1)}\right]^{-1} .$$

$$\begin{aligned} \ln \Lambda(\boldsymbol{d} \mid \boldsymbol{\vartheta}) &= \ln p(\boldsymbol{d} \mid \boldsymbol{\vartheta}, \boldsymbol{\mathcal{H}}_1) - \ln p(\boldsymbol{d} \mid \boldsymbol{\mathcal{H}}_0) \\ &= -\frac{1}{2}(\boldsymbol{d} - \boldsymbol{h}_{\boldsymbol{\vartheta}} \mid \boldsymbol{d} - \boldsymbol{h}_{\boldsymbol{\vartheta}}) + \frac{1}{2}(\boldsymbol{d} \mid \boldsymbol{d}) \\ &= (\boldsymbol{d} \mid \boldsymbol{h}_{\boldsymbol{\vartheta}}) - \frac{1}{2}(\boldsymbol{h}_{\boldsymbol{\vartheta}} \mid \boldsymbol{h}_{\boldsymbol{\vartheta}}) , \end{aligned}$$

选择合适的T,
[f_min,f_max] ,
[f_start, f_final]

至此，统计模型基本完备，还差PSD估计和波形模版

相关功率谱估计（Blackman-Tukey法）

$$\widehat{R}[k] = \frac{1}{N} \sum_{n=0}^{N-1-|k|} x[n+|k|]x^*[n],$$

$$\boxed{\widehat{S}_{\mathrm{cor}}(f) = \sum_{k} \widehat{R}[k] e^{-i2\pi fk/f_s}}$$

问题：

随着时延|k|增大，可用于平均的数据对越来越少，因此：
- 大时延处的相关函数估计波动很大；
- 通常要截断相关函数，使频谱被平滑，降低频率分辨率。

周期图（Periodgram）

$$x(n)=\begin{cases}x_N(n), & n=0\sim N-1,\\ 0, & \text{其它的 } n,\end{cases}$$

这时有

$$\hat{r}(k)=\frac{1}{N}\sum_{n=-\infty}^{\infty}x(n)x(n+k),$$

代入(6.1)式有

$$\begin{aligned}\hat{S}(\omega_j)&=\frac{1}{N}\sum_{k=-\infty}^{\infty}\sum_{n=-\infty}^{\infty}x(n)x(n+k)\mathrm{e}^{-\mathrm{i}k\omega_j}\\&=\frac{1}{N}\sum_{n=-\infty}^{\infty}x(n)\mathrm{e}^{\mathrm{i}n\omega_j}\sum_{k=-\infty}^{\infty}x(n+k)\mathrm{e}^{-\mathrm{i}(n+k)\omega_j}\\&=\frac{1}{N}[X'(\omega_j)]^*[X'(\omega_j)]\\&=\frac{1}{N}|X'(\omega_j)|^2,\end{aligned}$$

其中

$$\begin{aligned}X'(\omega_j)&=\sum_{n=-\infty}^{\infty}x(n)\mathrm{e}^{-\mathrm{i}n\omega_j}\\&=\sum_{n=0}^{N-1}x_N(n)\mathrm{e}^{-\mathrm{i}n\omega_j}\\&=X_N(\omega_j)=X_N(j),\end{aligned}$$

直接FFT再取模平方

问题:

1、泄漏效应严重
2、单次周期图随机起伏很大

加权交叠平均法（Welch方法）

原始数据

分成若干重叠数据段

[————————]

[————————]

[————————]

[————————]

$$X_\ell[k] = \sum_{n=0}^{M-1} w[n] x_\ell[n] e^{-i2\pi kn/M},$$

$$\widehat{S}_\ell(f_k) = \frac{|X_\ell[k]|^2}{f_s \sum_n w^2[n]}.$$

每段加窗、FFT、取模平方

↓ ↓ ↓ ↓

$P_1(f)$ $P_2(f)$ $P_3(f)$ $P_4(f)$

$$\widehat{S}_{\text{Welch}}(f_k) = \frac{1}{K} \sum_{\ell=1}^{K} \widehat{S}_\ell(f_k)$$

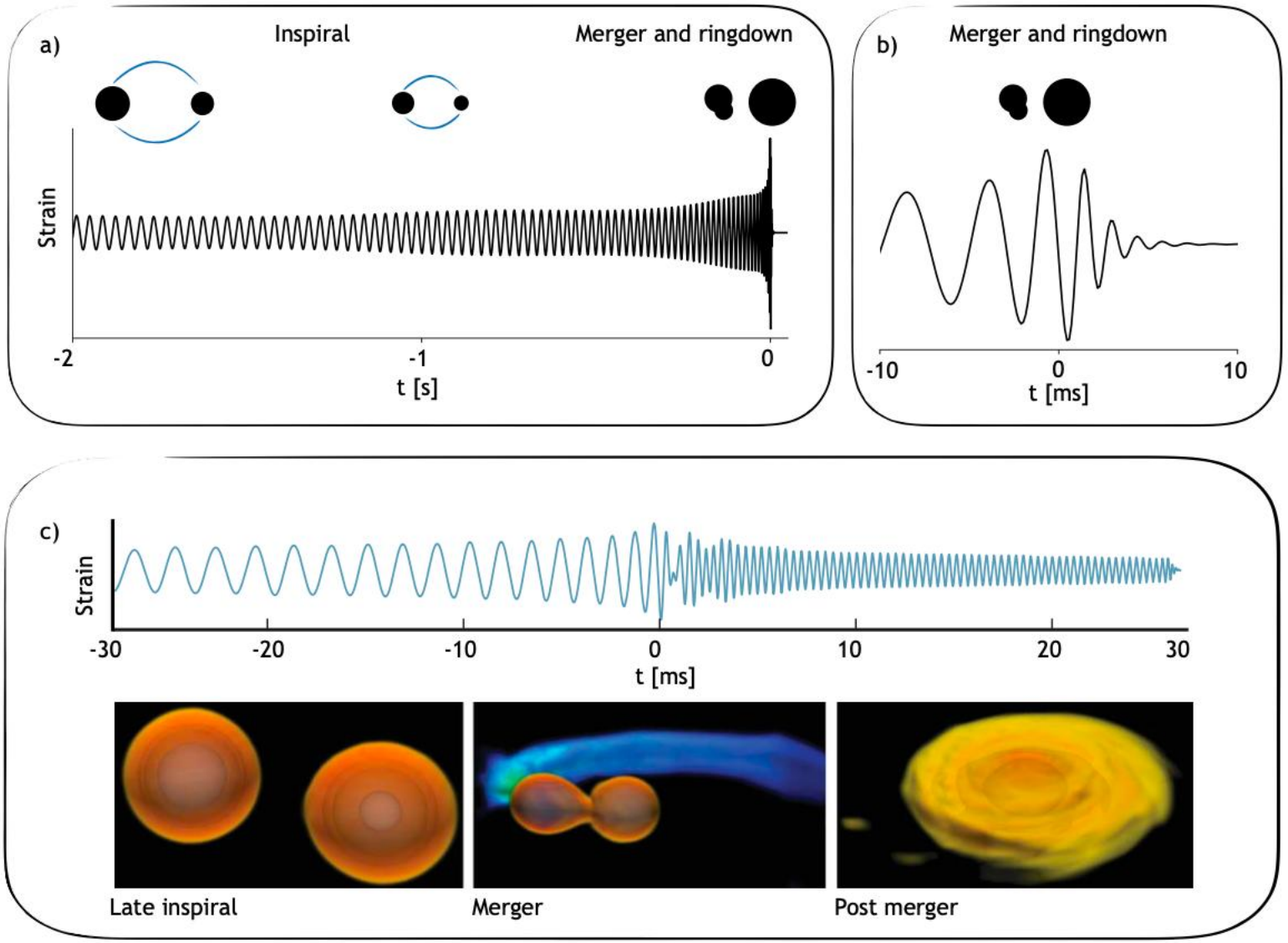
最后平均

$\hat{S}_{\text{Welch}}(f) = [P_1(f) + P_2(f) + P_3(f) + P_4(f)] / 4$

实际使用 median-mean

操作	目的	代价
分段	获得多个谱估计用于平均	每段更短，频率分辨率下降
加窗	减少首尾截断导致的泄漏	主瓣变宽
重叠	提高加窗后数据的利用率	各段不完全独立
平均	降低PSD估计的方差	可能平滑掉细小谱结构

波形模版


a)
Inspiral
Merger and ringdown
Strain
-2
-1
0
t [s]
b)
Merger and ringdown
-10
0
10
t [ms]
c)
Strain
-30
-20
-10
0
10
20
30
t [ms]
Late inspiral
Merger
Post merger

PhenomP一般取J-frame

LALSuite模版库

$$h_+(t) - \mathrm{i}h_\times(t) = \sum_{\ell=2}^{\infty} \sum_{m=-\ell}^{\ell} h_{\ell m}(t)\, {}_{-2}Y_{\ell m}(\theta_{\mathrm{JN}}, \phi)\,,$$

```
enum Approximant {
    TaylorT1 , TaylorT2 , TaylorT3 , TaylorF1 ,
    EccentricFD , TaylorF2 , TaylorF2Ecc , TaylorF2NLTides ,
    TaylorR2F4 , TaylorF2RedSpin , TaylorF2RedSpinTidal , PadeT1 ,
    PadeF1 , EOB , BCV , BCVSpin ,
    SpinTaylorT1 , SpinTaylorT2 , SpinTaylorT3 , SpinTaylorT4 ,
    SpinTaylorT5 , SpinTaylorF2 , SpinTaylorFrameless , SpinTaylor ,
    PhenSpinTaylor , PhenSpinTaylorRD , SpinQuadTaylor , FindChirpSP ,
    FindChirpPTF , GeneratePPN , BCVC , FrameFile ,
    AmpCorPPN , NumRel , NumRelNinja2 , Eccentricity ,
    EOBNR , EOBNRv2 , EOBNRv2HM , EOBNRv2_ROM ,
    EOBNRv2HM_ROM , TEOBResum_ROM , SEOBNRv1 , SEOBNRv2 ,
    SEOBNRv2_opt , SEOBNRv3 , SEOBNRv3_pert , SEOBNRv3_opt ,
    SEOBNRv3_opt_rk4 , SEOBNRv4 , SEOBNRv4_opt , SEOBNRv4P ,
    SEOBNRv4PHM , SEOBNRv2T , SEOBNRv4T , SEOBNRv1_ROM_EffectiveSpin ,
    SEOBNRv1_ROM_DoubleSpin , SEOBNRv2_ROM_EffectiveSpin , SEOBNRv2_ROM_DoubleSpin , SEOBNRv2_ROM_DoubleSpin_HI ,
    Lackey_Tidal_2013_SEOBNRv2_ROM , SEOBNRv4_ROM , SEOBNRv4HM_ROM , SEOBNRv4_ROM_NRTidal ,
    SEOBNRv4_ROM_NRTidalv2 , SEOBNRv4_ROM_NRTidalv2_NSBH , SEOBNRv4T_surrogate , HGimri ,
    IMRPhenomA , IMRPhenomB , IMRPhenomFA , IMRPhenomFB ,
    IMRPhenomC , IMRPhenomD , IMRPhenomD_NRTidal , IMRPhenomD_NRTidalv2 ,
    IMRPhenomNSBH , IMRPhenomHM , IMRPhenomP , IMRPhenomPv2 ,
    IMRPhenomPv2_NRTidal , IMRPhenomPv2_NRTidalv2 , IMRPhenomFC , TaylorEt ,
    TaylorT4 , EccentricTD , TaylorN , SpinTaylorT4Fourier ,
    SpinTaylorT5Fourier , SpinDominatedWf , NR_hdf5 , NRSur4d2s ,
    NRSur7dq2 , NRSur7dq4 , SEOBNRv4HM , NRHybSur3dq8 ,
    IMRPhenomXAS , IMRPhenomXHM , IMRPhenomPv3 , IMRPhenomPv3HM ,
    IMRPhenomXP , IMRPhenomXPHM , TEOBResumS , IMRPhenomT ,
    IMRPhenomTHM , IMRPhenomTP , IMRPhenomTPHM , SEOBNRv5_ROM ,
    SEOBNRv4HM_PA , pSEOBNRv4HM_PA , IMRPhenomXAS_NRTidalv2 , IMRPhenomXP_NRTidalv2 ,
    IMRPhenomXO4a , ExternalPython , SEOBNRv5HM_ROM , IMRPhenomXAS_NRTidalv3 ,
    IMRPhenomXP_NRTidalv3 , SEOBNRv5_ROM_NRTidalv3 , IMRPhenomXPNR , NumApproximants
}
```

Enum that specifies the PN approximant to be used in computing the waveform. More...

- **IMR**：Inspiral–Merger–Ringdown；
- **Phenom**：现象学拟合；
- **SEOBNR**：Spin Effective-One-Body Numerical Relativity；
- **AS**：aligned spin；
- **P**：precession；
- **HM**：higher modes；
- **PHM**：precession + higher modes；
- **T / NRTidal**：潮汐；
- **ROM**：reduced-order model；
- **Sur**：surrogate；
- **X、v4、v5**：模型世代，不是 PN 阶数。

https://lscsoft.docs.ligo.org/lalsuite/7.26/lalsimulation/group___l_a_l_sim_inspiral__h.html

Newtonian 波形

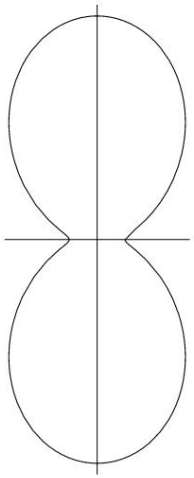
多极矩展开

弱场低速，波长大于线度

$$[h_{ij}^{TT}(t,\boldsymbol{x})]_{\rm quad} = \frac{1}{r}\frac{2G}{c^4}\Lambda_{ij,kl}(\hat{\boldsymbol{n}})\ddot{Q}_{kl}(t-r/c)$$

$$h_+(t) = \frac{4}{r}\left(\frac{GM_c}{c^2}\right)^{5/3}\left(\frac{\pi f_{\rm gw}}{c}\right)^{2/3}\frac{1+\cos^2\theta}{2}\cos(2\pi f_{\rm gw}t_{\rm ret}+2\phi)\,,$$

$$h_\times(t) = \frac{4}{r}\left(\frac{GM_c}{c^2}\right)^{5/3}\left(\frac{\pi f_{\rm gw}}{c}\right)^{2/3}\cos\theta\,\sin(2\pi f_{\rm gw}t_{\rm ret}+2\phi)\,,$$



$$f_{\rm gw}(\tau) \simeq 134\,{\rm Hz}\left(\frac{1.21M_\odot}{M_c}\right)^{5/8}\left(\frac{1\,s}{\tau}\right)^{3/8} \qquad M_c = \mu^{3/5}m^{2/5} = \frac{(m_1m_2)^{3/5}}{(m_1+m_2)^{1/5}}$$

$$\frac{dP}{d\Omega} = \frac{2}{\pi}\frac{c^5}{G}\left(\frac{GM_c\omega_{\rm gw}}{2c^3}\right)^{10/3}g(\theta) \qquad g(\theta) = \left(\frac{1+\cos^2\theta}{2}\right)^2+\cos^2\theta$$

木星-太阳：约几千瓦　　双黑洞：每秒一个太阳

频域模版

绝热近似：轨道参数缓慢漂移

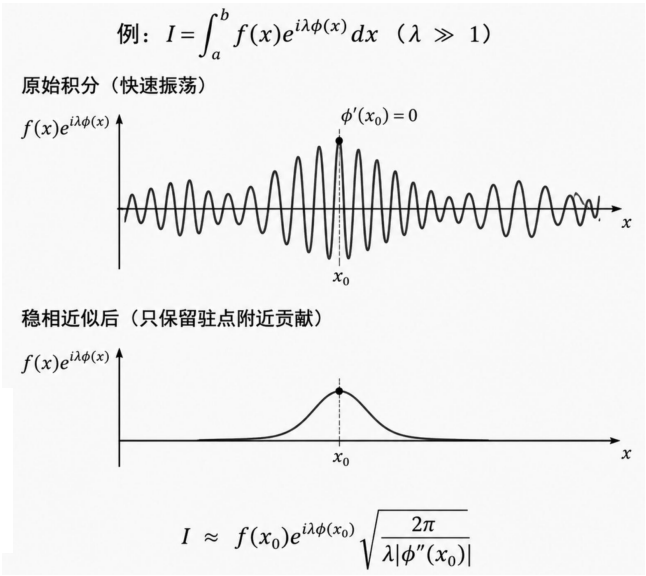
稳相近似：频域中频率f的主要贡献，来自时域信号的瞬时频率恰好经过f的时刻

$$\tilde{h}_+(f) = \frac{1}{\pi^{2/3}}\left(\frac{5}{24}\right)^{1/2} e^{i\Psi_+(f)} \frac{c}{d_L}\left(\frac{G\mathcal{M}}{c^3}\right)^{5/6} \frac{1}{f^{7/6}}\left(\frac{1+\cos^2\iota}{2}\right),$$

$$\tilde{h}_\times(f) = \frac{1}{\pi^{2/3}}\left(\frac{5}{24}\right)^{1/2} e^{i\Psi_\times(f)} \frac{c}{d_L}\left(\frac{G\mathcal{M}}{c^3}\right)^{5/6} \frac{1}{f^{7/6}}\cos\iota,$$

$$\Psi_+(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \frac{3}{4}\left(\frac{G\mathcal{M}}{c^3} 8\pi f\right)^{-5/3}$$

$$\dot{f}_{\rm gw} = \frac{96}{5}\pi^{8/3}\left(\frac{GM_c}{c^3}\right)^{5/3} f_{\rm gw}^{11/3}. \qquad \boxed{\tilde{h}(f) = \mathcal{A} f^{-7/6} e^{i\Psi(f)}}.$$



高阶效应

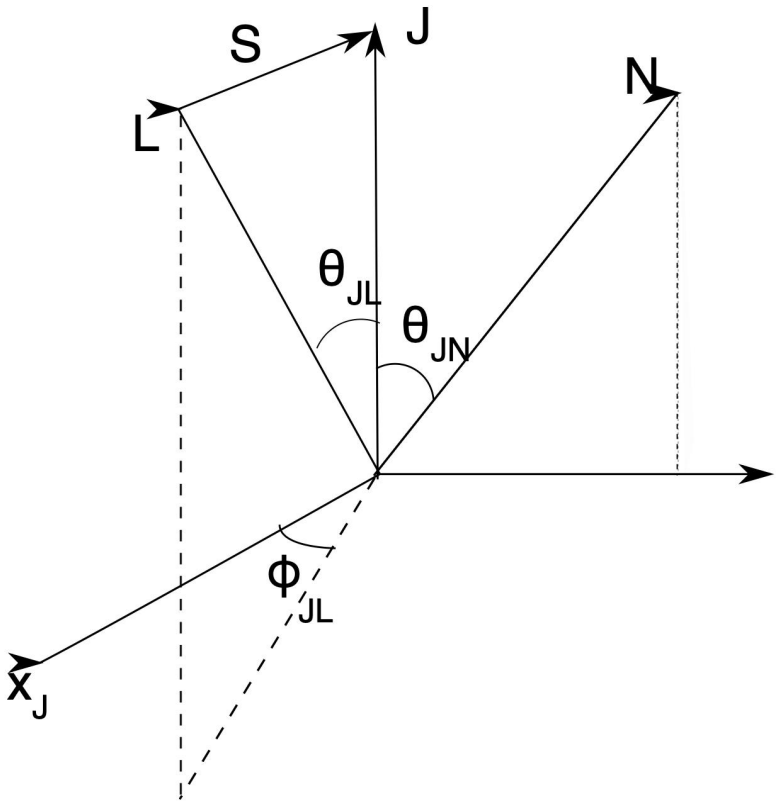
$$\Psi_{\rm SPA}(f) = 2\pi f t_c - \phi_c + \frac{3}{128\eta v^5}\left[\Psi_{\rm NS} + v^3\Psi_{\rm SO} + v^4\Psi_{\rm SS} + v^7\Psi_{\rm SSS} + \mathcal{O}(v^8)\right] + \Psi_{\rm H}$$

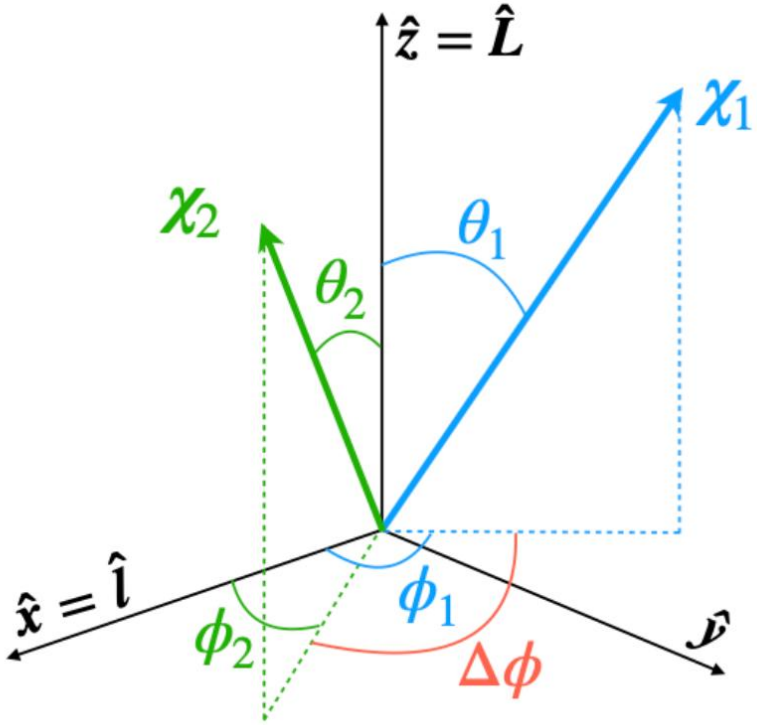
内禀参数与外部参数

参数	中文名称	常用符号
Right ascension (R.A.)	赤经	α
Declination (Decl.)	赤纬	δ
Luminosity distance	光度距离	d_L
Inclination angle	倾角②	θ_{JN}
Polarization angle	极化角	ψ
Geocentric Coalescence time	并合时间	t_c
Coalescence phase	并合相位	ϕ_c
Chirp mass	啁啾质量	$\mathcal{M}$
Mass ratio	质量比	$q(<1)$
Spin magnitude	自旋大小	a_i
Spin tilt angle	自旋倾角③	t_i
Spin azimuthal angle	自旋夹角④	$\Delta\phi$
Azimuthal position of $\hat{L}$	轨道进动角⑤	ϕ_{JL}
Tidal deformability	潮汐形变度⑥	Λ_i

注意点：参考频率，redshifted质量

自旋参数表示

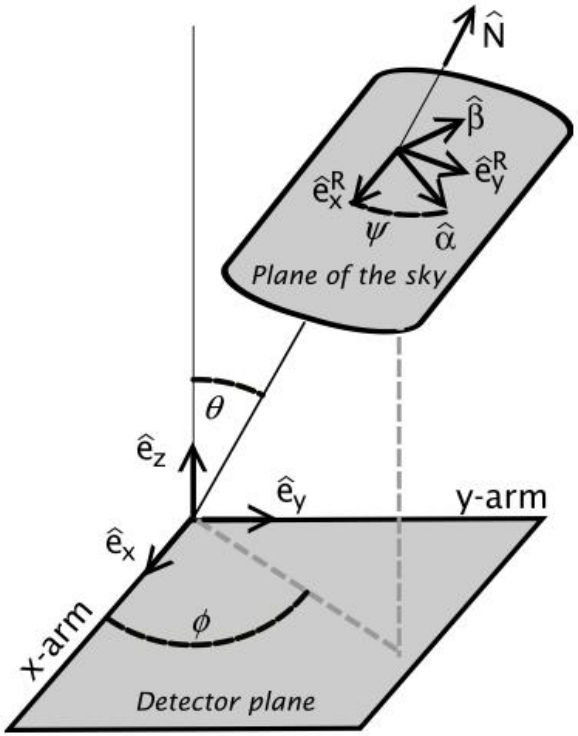




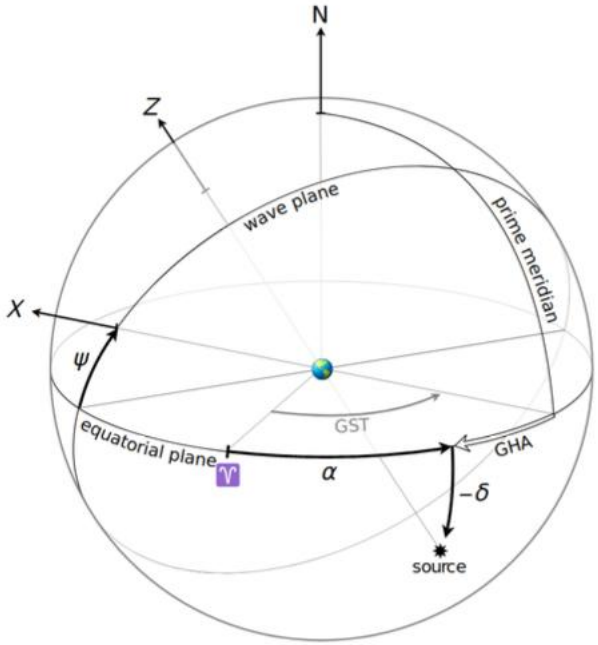
优点：

1. 保守2PN以下，J相对稳定
2. 参数估计时，采样器友好

探测器响应函数



X一般约定取轨道角动量在天球切面上的投影方向

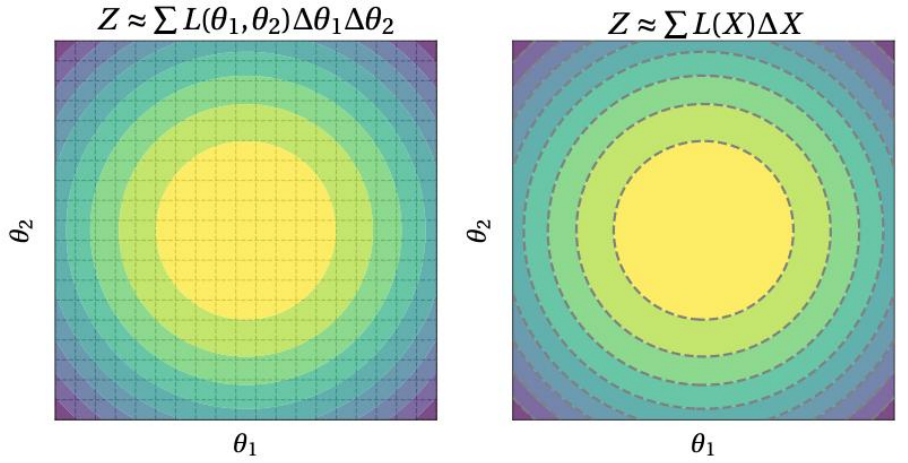


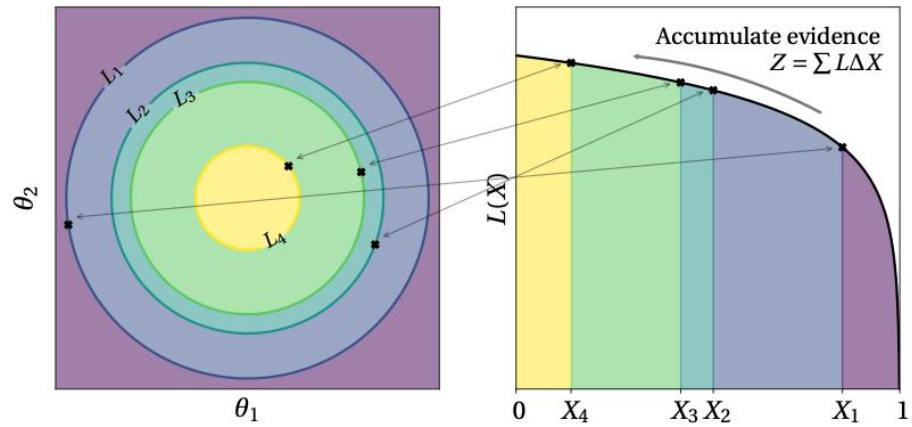
$$F_+ = \frac{1}{2}(1+\cos^2\theta)\cos 2\phi\cos 2\psi - \cos\theta\sin 2\phi\sin 2\psi$$

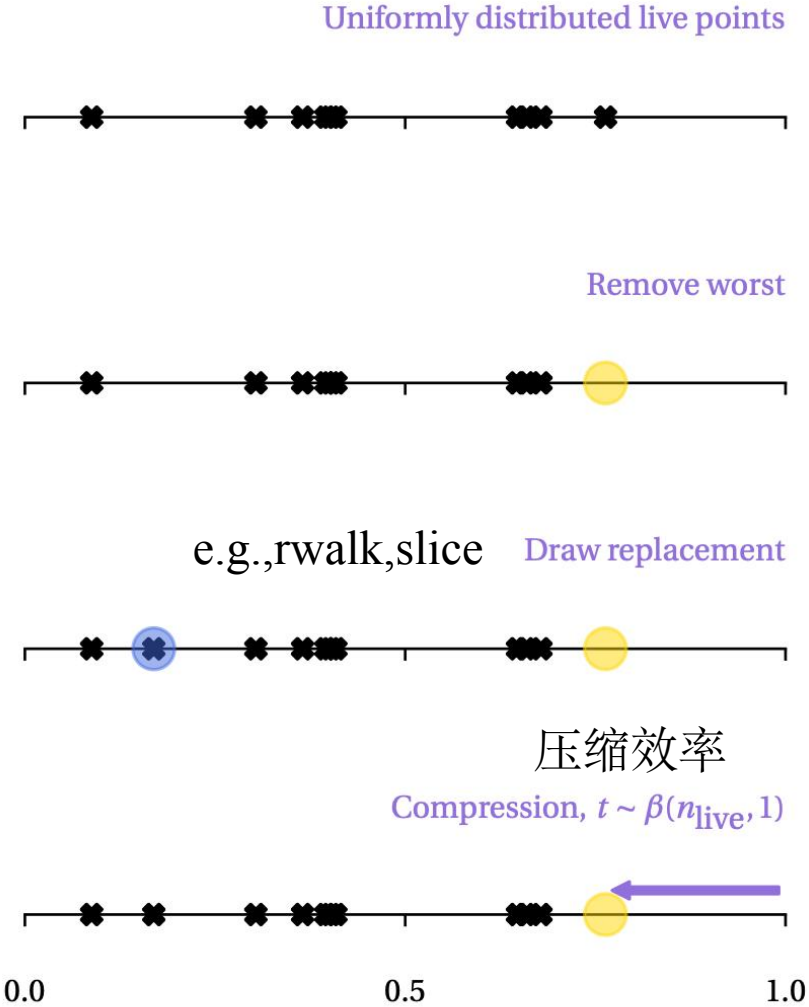
$$F_\times = \frac{1}{2}(1+\cos^2\theta)\cos 2\phi\sin 2\psi + \cos\theta\sin 2\phi\cos 2\psi$$

$$h^{\mathrm{I}}(t) = F_+^{\mathrm{I}}(\alpha,\delta,\psi,t_c)h_+(\theta_{\mathrm{GW}},\delta t^{\mathrm{I}},t) + F_\times^{\mathrm{I}}(\alpha,\delta,\psi,t_c)h_\times(\theta_{\mathrm{GW}},\delta t^{\mathrm{I}},t)$$

嵌套采样







$$\Delta \log Z \propto 1/\sqrt{n_{\text{live}}}$$

$$\texttt{dlogZ} \lesssim \Delta \log Z$$

$$\text{Speed-up} \approx n_{\text{live}} \log\left(1 + \frac{n_{\text{CPU}}}{n_{\text{live}}}\right)$$

$$N_{\text{live}} \sim \max\left(cD, \frac{D(D+1)}{2}\right)$$

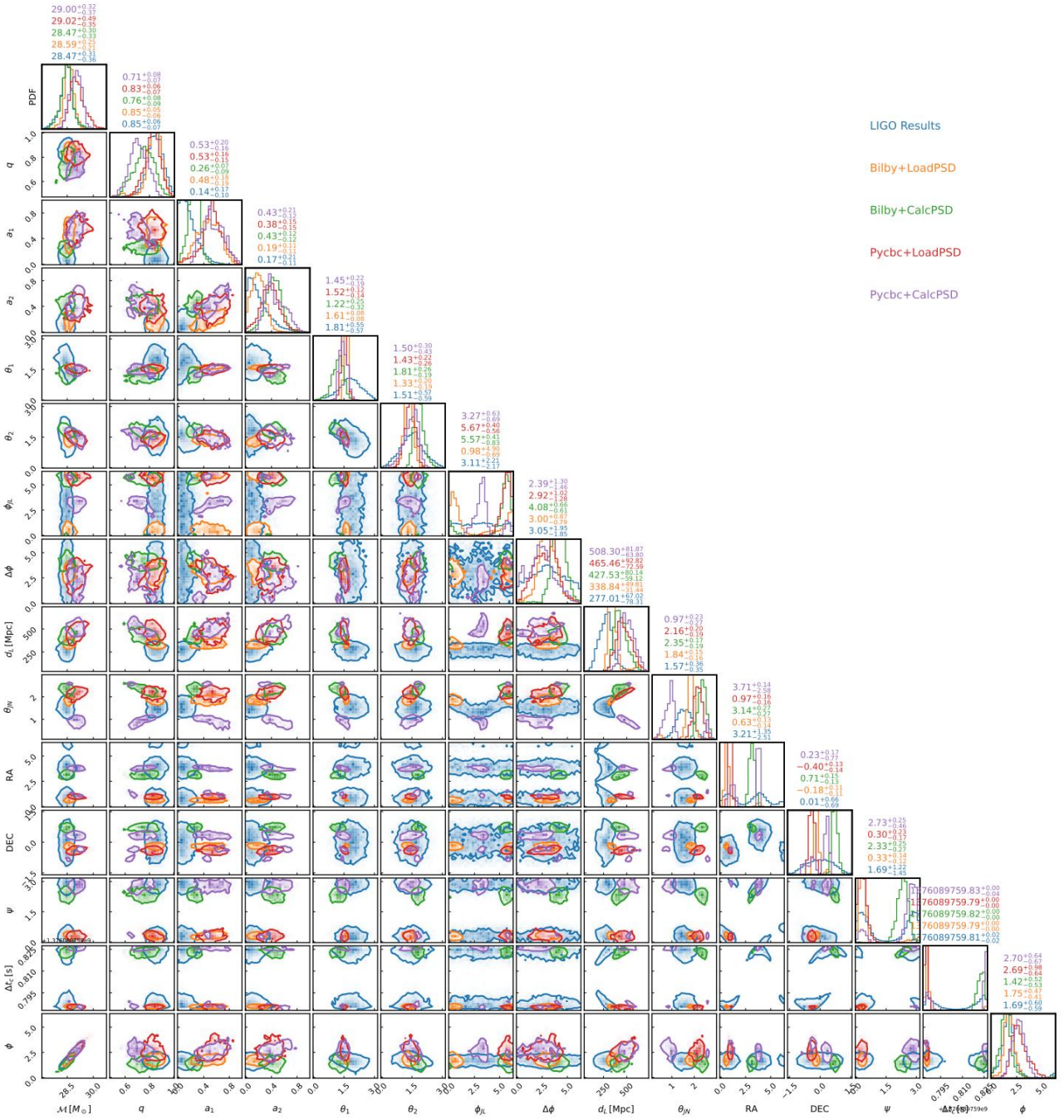
acceptance rate

选择采样器

Code	Methods	Dynamic	Languages	Field	Pub. Year
CosmoNest [60, 61]	ellipsoid	fixed	Fortran	Cosmology	2006
MultiNest [48, 84]	multi-ellipsoid	fixed	Fortran, C/C++, Python	Cosmology	2008
DIAMONDS [249]	multi-ellipsoid	fixed	C++	Astrophysics	2015
nestle [250]	ellipsoid, multi-ellipsoid	fixed	Python	Astrophysics	2015
nessai [90, 91]	normalising flow ellipsoid	fixed	Python	Gravitational waves	2021
(dy)PolyChord [53, 65]	slice	dynamic	Fortran, C/C++, Python	Cosmology	2015
LALInferenceNest [180]	random walk, ensemble, differential evolution	fixed	C	Gravitational waves	2015
Nested_fit [104, 257, 258]	random walk	fixed	Fortran	Atomic physics	2016
cpnest [259]	slice, differential evolution, Gauss, Hamiltonian, ensemble	fixed	Python	Gravitational waves	2017
pymatnest [44]	random walk, Galilean, symplectic Hamiltonian	fixed	Python	Materials	2017
NNest [261]	normalising flow random walk	fixed	Python	Cosmology	2019
DNest5 [55]	user-defined, random walk	diffusive	C++	Astrophysics	2020
BayesicFitting [263]	random walk, slice, Galilean, Gibbs	fixed	Python	Astronomy	2021
dynesty [52]	ellipsoid, multi-ellipsoid, MLFriends & Gauss, slice, Hamiltonian	dynamic	Python	Astrophysics	2020
UltraNest [92]	MLFriends + ellipsoid & Gauss, hit-and-run, slice	reactive	Python, Julia, R, C/C++, Fortran	Astrophysics	2020
jaxns [266]	multi-ellipsoid & slice	fixed	jax	Astronomy	2021

边缘化一些外禀参数，机器学习（Normalizing flow）

结果示意

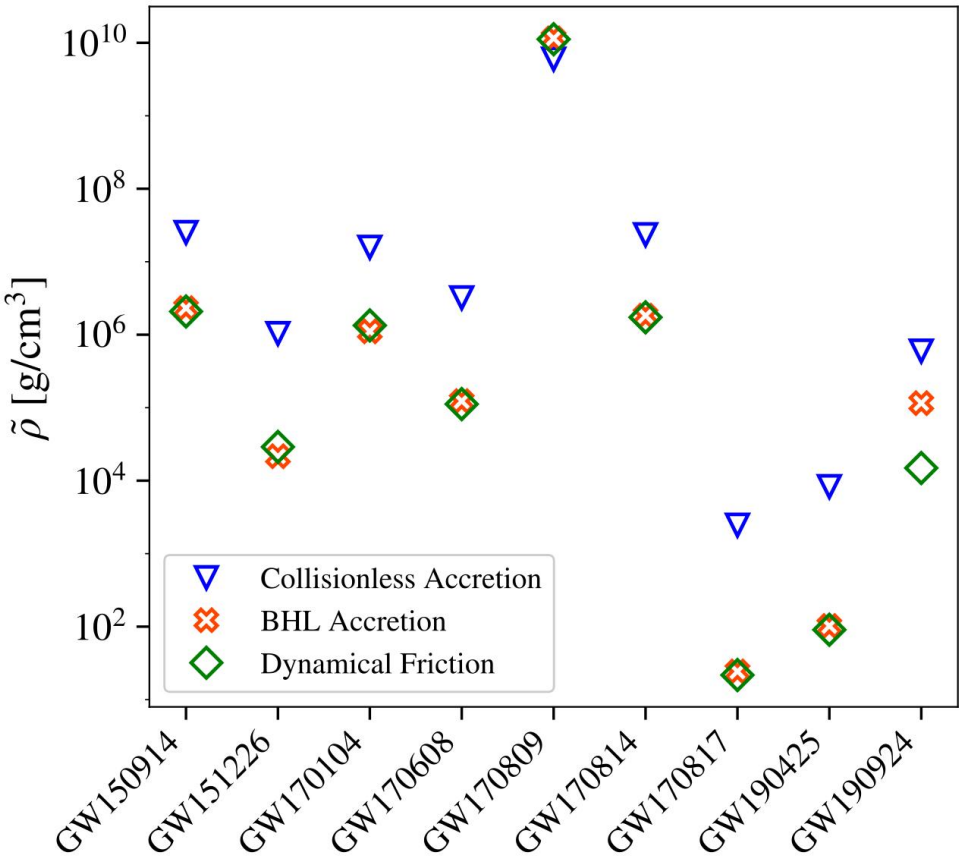

LIGO Results
Bilby+LoadPSD
Bilby+CalcPSD
Pycbc+LoadPSD
Pycbc+CalcPSD
PDF
$\mathcal{M}\,[M_\odot]$
q
a_1
a_2
θ_1
θ_2
ϕ_{JL}
$\Delta\phi$
d_L [Mpc]
θ_{JN}
RA
DEC
ψ
Δt_c [s]
ϕ

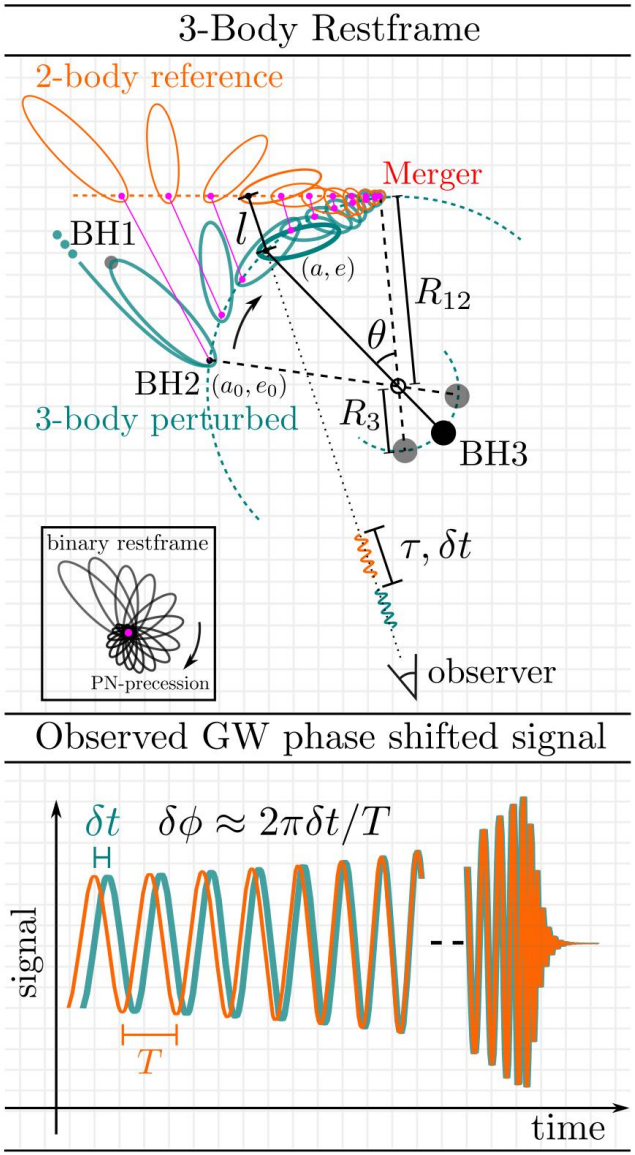
可能的应用：

AGN候选源检验

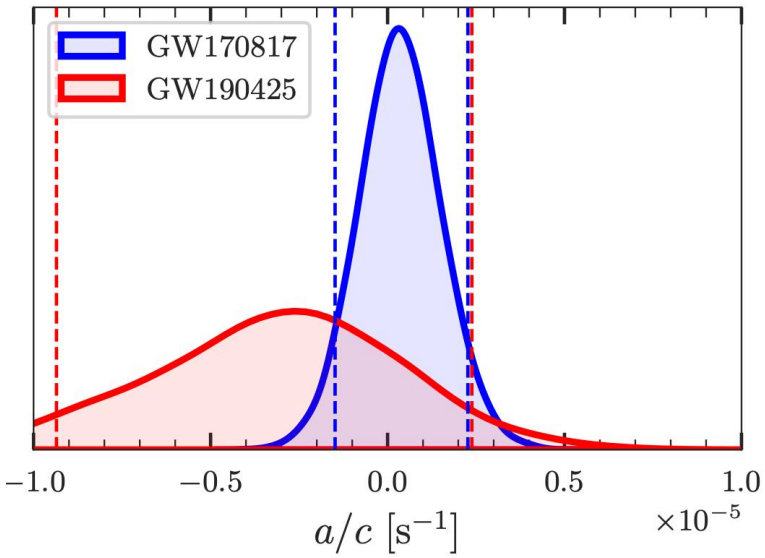
$$\phi^{\rm env} = \phi^{\rm vac} + \frac{3}{128\eta}\delta\Phi_k v^{k-5} \qquad \delta\Phi_k = -\beta_k \tilde{\rho} M^2$$

Effect	β_k	k
CA	$\frac{125\pi(1-3\eta)}{357\eta^2}$	-9
BHLA	$\frac{125\pi[1-5\eta(1-\eta)]}{1824\eta^4}$	-11
DF	$\frac{25\pi(1-3\eta)}{304\eta^3}$	-11





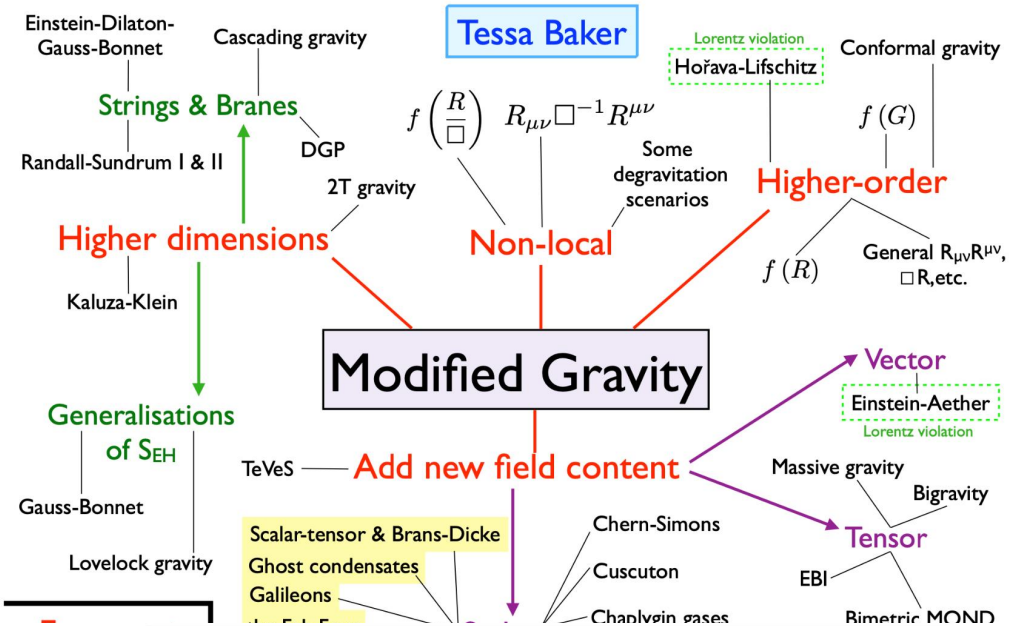
$$\Delta\Psi(f) = \frac{25}{65536\,\eta^2}\left(\frac{GM}{c^3}\right)\left(\frac{a}{c}\right)v_f{}^{-13}$$

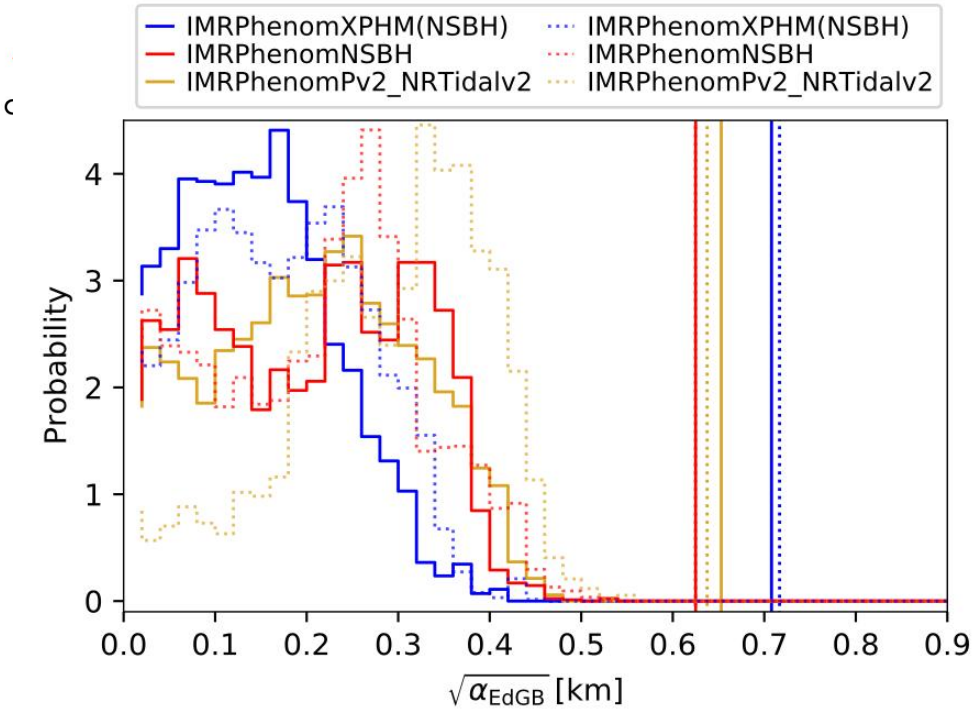


$$\tilde{h}(f) = \tilde{h}_{\rm GR}(1 + \alpha\, u^a) e^{i\delta\Psi}$$

$$u = (\pi \mathcal{M} f)^{\frac{1}{3}} \qquad \delta\Psi = \beta u^b$$

Theories	PPE Phase Parameters		Binary Type
	Magnitude (β)	Exp. (b)	
Scalar-Tensor [95, 96]	$-\frac{5}{7168}\eta^{2/5}(\alpha_1 - \alpha_2)^2$	-7	Any
EdGB [97]	$-\frac{5}{7168}\zeta_{\rm EdGB}\frac{\left(m_1^2 \tilde{s}_2^{\rm EdGB} - m_2^2 \tilde{s}_1^{\rm EdGB}\right)^2}{m^4 \eta^{18/5}}$	-7	Any
DCS [82, 98]	$\frac{481525}{3670016}\eta^{-14/5}\zeta_{\rm dCS}\left[-2\delta_m \chi_a \chi_s + \left(1 - \frac{4992\eta}{19261}\right)\chi_a^2 + \left(1 - \frac{72052\eta}{19261}\right)\chi_s^2\right]$	-1	BH/BH
Einstein-Æther [99]	$-\frac{5}{3584}\eta^{2/5}\frac{(s_1^{\rm EA} - s_2^{\rm EA})^2}{[(1-s_1^{\rm EA})(1-s_2^{\rm EA})]^{4/3}}\left[\frac{(c_{14}-2)w_0^3 - w_1^3}{c_{14} w_0^3 w_1^3}\right]$	-7	Any
Khronometric [99]	$-\frac{5}{3584}\eta^{2/5}\frac{(s_1^{\rm kh} - s_2^{\rm kh})^2}{[(1-s_1^{\rm kh})(1-s_2^{\rm kh})]^{4/3}}\sqrt{\bar{\alpha}_{\rm kh}}\left[\frac{(\bar{\beta}_{\rm kh}-1)(2+\bar{\beta}_{\rm kh}+3\bar{\lambda}_{\rm kh})}{(\bar{\alpha}_{\rm kh}-2)(\bar{\beta}_{\rm kh}+\bar{\lambda}_{\rm kh})}\right]^{3/2}$	-7	Any
Noncommutative [100]	$-\frac{75}{256}\eta^{-4/5}(2\eta - 1)\Lambda^2$	-1	BH/BH
Varying-G [92]	$-\frac{\mathbf{25}}{\mathbf{851968}}\eta_0^{\mathbf{3/5}}\dot{\mathbf{G}}_{\mathbf{C},0}\left[\mathbf{11 m_0 + 3(s_{1,0} + s_{2,0} - \delta_{\dot{G}})m_0 - 41(m_{1,0}s_{1,0} + m_{2,0}s_{2,0})}\right]$	$\mathbf{-13}$	Any


Tessa Baker
Modified Gravity
Strings & Branes
Einstein-Dilaton-Gauss-Bonnet
Cascading gravity
Randall-Sundrum I & II
DGP
Higher dimensions
Kaluza-Klein
2T gravity
Non-local
$f\left(\frac{R}{\Box}\right)$
$R_{\mu\nu}\Box^{-1}R^{\mu\nu}$
Some degravitation scenarios
Higher-order
Lorentz violation
Hořava-Lifschitz
Conformal gravity
$f(G)$
$f(R)$
General $R_{\mu\nu}R^{\mu\nu}$, $\Box R$,etc.
Generalisations of S_{EH}
Gauss-Bonnet
Lovelock gravity
Add new field content
TeVeS
Scalar-tensor & Brans-Dicke
Ghost condensates
Galileons
Chern-Simons
Cuscuton
Vector
Einstein-Aether
Lorentz violation
Massive gravity
Bigravity
Tensor
EBI
Bimetric MOND


IMRPhenomXPHM(NSBH)
IMRPhenomNSBH
IMRPhenomPv2_NRTidalv2
IMRPhenomXPHM(NSBH)
IMRPhenomNSBH
IMRPhenomPv2_NRTidalv2
Probability
0
1
2
3
4
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
$\sqrt{\alpha_{\mathrm{EdGB}}}$ [km]

可能的方向：

短时标和超长时标信号的处理

机器学习应用于参数估计、模版构造